

Project Initialization and Planning Phase

Date	03 July 2026
Team ID	N/A
Project Title	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

OptiCrop applies machine learning to agricultural decision-making, helping farmers choose the right crop for their soil and climate conditions. It tackles inefficient crop selection and resource waste, aiming for higher yields, better resource use, and more confident farming and policy decisions. Key features include a multi-algorithm crop recommendation model and a real-time Flask web interface.

Project Overview	
Objective	Help farmers, researchers, and policymakers make faster, more accurate crop-selection decisions by applying machine learning to soil nutrient (N, P, K) and climate (temperature, humidity, pH, rainfall) data.
Scope	Covers data collection, exploratory data analysis, preprocessing (null/outlier handling, seasonal grouping), training and comparing multiple ML algorithms, and deploying the best model through a Flask web application for real-time crop recommendations.
Problem Statement	
Description	Farmers often select crops based on tradition or guesswork rather than actual soil nutrient and climate conditions, leading to mismatched crop choices, lower yields, and wasted water/fertilizer.
Impact	Better-matched crop selection increases yield and farmer income, reduces wasted resources and environmental impact, and gives researchers/policymakers reliable, data-backed insight into regional crop suitability.
Proposed Solution	
Approach	Train and compare multiple supervised classification algorithms (KNN, Logistic Regression, Decision Tree, Random Forest) on N, P, K, temperature, humidity, pH, and rainfall to predict the most suitable crop; K-Means clustering is used exploratively to identify natural groupings of agricultural conditions.
Key Features	- ML-based crop recommendation from 7 soil/climate parameters. - Comparison of 4 classification algorithms using Accuracy, Precision, Recall, and F1-Score. - Best-performing model (Random Forest) persisted with Pickle for instant reuse without retraining. - Simple Flask web app (Home, About, Find Your Crop) for real-time predictions. - Season-aware crop grouping (Summer/Winter/Rainy) to add environmental context.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU, 2+ cores (no GPU required — classical ML models only)
Memory	RAM specifications	4 GB
Storage	Disk space for data, models, and logs	< 500 MB (dataset + model/scaler files)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm, Anaconda Navigator
Data		
Data	Source, size, format	Representative crop-recommendation dataset (2,200 records, 22 crop classes, csv), structured like Kaggle's Smart Agricultural Production Optimizing Engine dataset (N, P, K, temperature, humidity, ph, rainfall, label)